

MURUGESAN M

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EDUCATION

Madras Institute of Technology, Anna University B.E. Electronics and Communication Engineering CGPA: 8.16 (up to 4th semester)	Chennai, Tamil Nadu 2023 – 2027
St. Antony's Matric Higher Secondary School HSC-95.17%	Tuticorin, Tamil Nadu May 2023

COURSEWORK

- Digital Electronics and Analog System Design, Computer Architecture and Organization, Control Systems
- Micro controllers, Communication Networks, Transmission Lines, Antenna Wave Propagation, Digital Communication

SKILLS

Programming

C, C++, Data Structures, Python, MATLAB (500+ lines of coding experience)

Tools

LTspice, Proteus, FOSSEE eSim (PCB Design), Altium Designer, Cisco Packet Tracer, CST studio, Xilinx Vivado, Cadence

LEADERSHIP & ACTIVITIES

Social Media Manager Event Organizer Event Lead – Quantumit Club, Chakravyuha'25, MIT Chennai	August 2025
Managing the social media posts and videos and edits publishing and advertising- Lead planning, coordination, logistics, team management, and public communication	
Volunteer – National Service Scheme (NSS) , MIT, Chennai	Present
Participated in community outreach, awareness rallies, and social initiatives	

PROJECTS

Traffic Surveillance System	April 2025
Developed a MATLAB-based system using YOLOv8 and Kalman Filter for real-time vehicle detection, tracking, and lane analysis, automotive safety systems - Integrated speed sensors to detect violations and alert authorities	
Overvoltage Protection System	April 2025
Designed surge detection with automatic power cutoff and safe recovery- Integrated temperature sensors and relays for short circuit fire hazard detection and alerts	
Multichannel Audio Streaming Optimization	April 2025
Implemented audio multiplexing with companding for efficient multi-signal transmission- Ensured high-quality reconstruction using compression and decompression	
Driver Drowsiness Detection System	Nov 2024
Designed a real-time driver alert system using IR sensors to monitor eye closure- Triggered alarms, braking control, System Integration and LED alerts to improve road safety	
IoT-Based Digital Letter Tracking and Management System	May 2026
Developed a QR-based IoT letter tracking system using Flask, ESP32, and PostgreSQL for real-time workflow monitoring and status updates. Implemented secure role-based dashboards, cloud database integration, and automated QR scanning for institutional document management.	
Wrist Pulse Signal Classification for Vata, Pitta, Kapha Detection	
Developed a wearable multi-sensor pulse acquisition system using Arduino and optical sensors for real-time physiological signal monitoring. Building a Python-based signal processing and ML dataset pipeline for Dosha classification using multi-channel pulse waveform analysis.	

EXPERIENCE

Embedded Systems Webinar

- gained exposure to embedded systems, electronics foundation, and embedded c coding

Hands on Workshop - RISC-V on FPGA

- gained exposure to RISC-V architecture and how to build circuits using VIVADO software on FPGA boards integrating and porting applications

RUSA Workshop on Embedded hardware validation through modern test equipment

- gained exposure to embedded systems, automotive applications, Robotics and social projects , Embedded systems calibration and validation and testing equipments- embedded system for RADAR, Power control , EDGE AI Systems
- LAB visit to IIT Madras gaining exposure to PCB Fabrication Process and Various Testing equipments for calibration , validation of modern systems

ACHIEVEMENTS

Runner-Up – Q-Debate, Chakravyuha’25

- Secured 2nd place in a technical debate on Classical vs Quantum Computing- Demonstrated strong critical thinking and technical depth

CERTIFICATIONS

- MATLAB Onramp and Linear Algebra – MathWorks
- IEEE Webinar: FPGA-Based Sensing for IoT (MNNIT Allahabad)
- NPTEL Certification – Python for Data Science